

The latest in the line up is probably French Bixi, a portable gesture device. It is a combination of Gorilla Glass and Bluetooth Low Energy (BLE) with optical sensing that can detect eight gestures.

Wearables and gesture recognition

Seen as the easiest way to break the stranglehold of keyboards and mice, gesture control through gloves is a promising area. Apotact Labs recently brought out a glove called Gest, which uses skeletal models and motion-processed data to sense gestures.

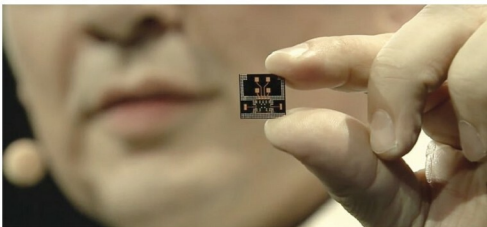
Essentially a glove-like controller, it comes with a software development kit (SDK) that allows engineers to build new applications for the platform. The glove packs 15 sensors including accelerometers, gyroscopes, magnetometers and more. The vendor claims that their software is smart enough to learn how you use your hands and uses that to create a personalised model.

Alternatively, as a more portable solution, Thalmic Labs brought out Myoarmband, which measures electrical activity in the forearm muscle to make sense of gestures. It lets you control drones or integrate these with Oculus Rift. However, the unique factor here is also its weakness, as there are many reports that the product is not ready for primetime yet.

Aria Wearable is more of a watch-like device from Deus Ex Technology, which will let you interact with electronics through simple finger gestures.

Funded by Menlo Ventures and Sequoia Capital, Nod probably uses the smallest form factor for a wearable gesture controller. It is a BLE-enabled gesture-control ring. It comes with highly-sensitive sensors built into the hardware that allows developers and engineers to make use of up to 12,600-dots-per-centimetre (32,000dpi) sensitivity.

Gesture-control rings have been



Ivan Poupyrev from Google ATAP holding a chip as he demonstrate an interaction sensor

Plug-n-Gesture

Cube camera-projector system is a plug-n-play style device that has the vision system and gesture-recognition software in place inside the box. This makes it easy to deploy in places with non-technical staff, where its camera-projector combo can detect the motion of people walking by.

in the works for a long time, including ones that did not really take off. Now we have Ring Zero from the firm Logbar. It is a product that was first seen at Kickstarter but is now available to all. Developers can access Ring SDK for iOS, but the one for Android is still a work-in-progress. The drawback here is that it requires the user to wear the sensing instrument.

Just smile and wave

3Gear Systems is a start-up pursuing the vision that gesture control should be implemented without having to touch any electronics in the first place. They are pursuing that vision through vision-enabled gesture recognition powered by a tiny camera-controlled device.

Leap Motion is another company into this same space, but they offer a gesture-recognition solution in a box. Their device can be used with devices like laptops and even Oculus Rift. It works by using a mix of two cameras and three infrared light emitting

diodes (LEDs). The cameras track light from the LEDs at a wavelength of 850 nanometers. This wavelength is outside the visible spectrum, so users cannot see it at all. Once this piece of hardware gets data, it passes it to Leap Motion tracking software that does all data crunching.

ArcSoft's technology enables gesture recognition on devices having stereoscopic or even single-lens devices. This enables using natural hand gestures as well as face, eye and finger motions to interact with devices. ArcSoft claims that their capabilities for hand gestures are highly accurate, detecting clear gestures at five metres even under low-light or back-lighting conditions.

Another example is Seoul based VTOUCH, which lets users control devices by tracking their eyes and fingers using a 3D camera.

AMD, too, is interested in this area, probably to strengthen the market for their accelerated processing units. AMD Gesture Control software is freely available, which allows users to utilise cameras such as their devices' built-in cameras as gesture-recognition tools. The company claims it delivers good accuracy in low-light conditions.

TedCas is a firm that specialises in building touch-free natural UI based medical environments. They focus on the use of optoelectronic devices to deliver intuitive interac-